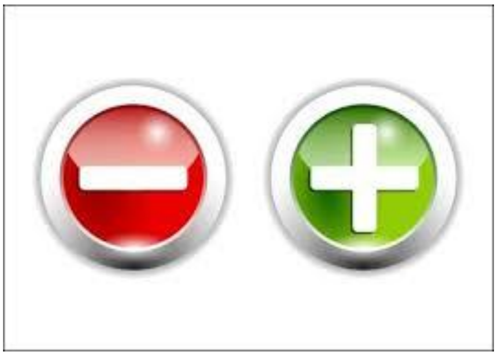
22PH202 & ELECTROMAGNETISM AND MODERN PHYSICS

UNIT I – ELECTRICITY

- Electric monopoles
- Electric field
- Electric flux
- Electric potential

Introduction

- Electricity is a form of energy that involves the movement of electrons from one point to another.
- It is used to heat things up, Example-Microwave, Toaster, Kettle
- It is used to give light and reduce darkness, Example- A tube light, lamp
- It is used to cool things, Example-Fridge
- A charged object exerts a force – a push or a pull
- There are two types of charges – negative and positive



- Electric forces, a type of non-contact force, can act on objects without touching. These forces are caused by electric charges in matter.
- Electric forces can attract or repel, depending on the charges involved.
- Here are some facts and characteristics of the electrostatic force.
- Like charges repel, and opposite charges attract
- Directly proportional to the product of two point-charges
- Inversely proportional to the distance of separation between the charges
- Acts along the line joining the two charges
- Friction: Rubbing two objects together can cause electrons to be “wiped” from one object and transferred to the other.
- **Conduction**: Transfer of electrons from one object to another through direct contact.
- Touching a negatively charged plastic ruler to an uncharged metal rod causes electrons from the ruler to travel to the rod. The rod becomes negatively charged by conduction.
- **Induction**: Occurs when charges in an uncharged object are rearranged without direct contact with the charged object
- A negatively charged balloon induces a positive charge on a small section of a wall because the electrons in the wall are repelled and move away from the balloon.
- **Static electricity** is an imbalance of electric charges within or on the surface of a material. The charge remains until it is able to move away by an electric current or electrical discharge.

- Rub a balloon on your hair on a dry day, Negative charges are transferred from the hair to the balloon, As the balloon is removed, it pulls on the hair with an attractive electric force.

Rubbing a balloon on your hair
Walking on a carpeted floor
Lightning
When a dry hair is brushed with a plastic comb

Examples for
Static Charges
or
Static Electricity

Dry Human Hands	More Positive
Rabbit Fur	
Glass	
Human Hair	
Fur	
Silk	
Paper	
Wood	
Amber	
Polyester	
Styrofoam	
Plastic Wrap	
PVC	
Silicon	
Teflon	More Negative

Electric Charges:

- Electric charge (symbol q) is the physical property of matter that causes it to experience a force when placed in an electromagnetic field.
- A coulomb (C) is the standard unit of electric charge in the International System of Units (SI). It is the amount of electricity that a 1-ampere (A) current carries in one second (s).
- A quantity of 1 C is equal to the electrical charge of approximately 6.24×10^{18} electrons.
- Examples: Glass rod rubbed with silk cloth, An ebonite rod rubber with woollen cloth.
- **Properties of Electric Charges:** Electric charge is an intrinsic property of matter
- Two basic charges (positive and negative). Opposite charges attract and Like charges repel
- Electric charge is always conserved. Quantization of charge, $q = \pm Ne$

COULOMB'S LAW STATEMENT

- Between Two Point Charges There Is Force Of *Attraction* Or *Repulsion* Depending Upon the ***Nature Of Charges.***

Charles Augustine de Coulomb

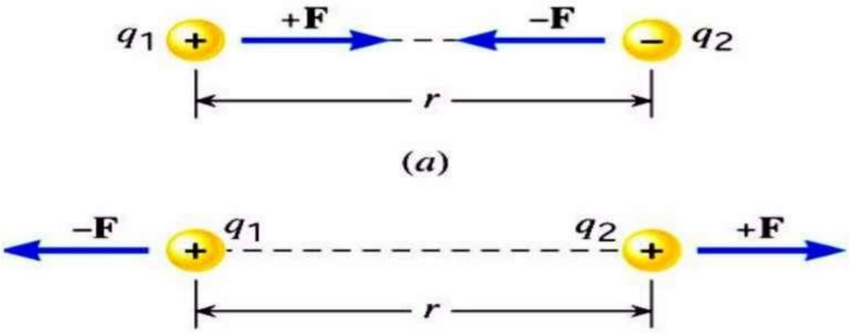


Fig. Understanding Of Attraction And Repulsion

- Coulomb Showed That:

1) $F \propto Q_1Q_2$

2) $F \propto \dfrac{1}{R^2}$

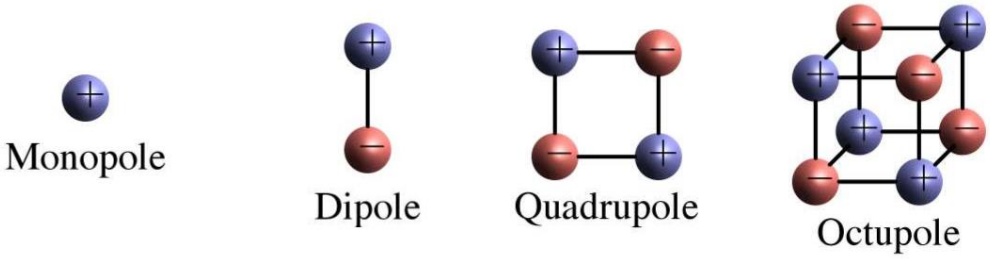
3) The Magnitude Of the Force Depends on the *Medium.*

- By Combining Two Equations We Get,

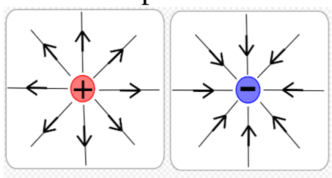
$$F \propto \frac{Q_1Q_2}{R^2}$$

Electric monopole

- A monopole is a single electric charge. A positive charge will have electric field lines going out from it while a negative charge will have electric field lines coming towards it.
- Electric monopoles exist in the form of particles that have a positive or negative electric charge, such as protons or electrons.
- An electric dipole is defined as a couple of opposite charges "q" and "–q" separated by a distance "d". By default, the direction of electric dipoles in space is always from negative charge "-q" to positive charge "q". The midpoint "q" and "–q" is called the centre of the dipole
- Quadrupole: It consists of two dipoles with dipole moments that are equal in magnitude but opposite in direction.
- Octopole is a system composed of eight electric charges arranged as four dipoles or two quadrupoles. The midpoint "q" and "–q" is called the centre of the dipole.



- **Electric filed lines;** An electric line of force is a smooth curve drawn in an electric field. The tangent at any point on the curve indicates the direction of the electric field at that point.



Nature of electric filed lines

- Electric field lines move away from positive charge and move towards negative charge
- Electric filed lines are conservative in nature
- Electric filed lines start from positive charge and terminate at negative charge. Also, field lines never cross each other.
- If there is only one charge, then electric filed terminate at infinity.

- **Electric field**: a region around a charged particle or object within which a force would be exerted on other charged particles or objects.
- **Summary**
- An electric dipole consists of two equal and opposite charges (q and -q) separated a distance d.
- Electric field E at any point is defined in terms of the electric force F that acts on a small positive test charge placed at that point divided by the magnitude q_0 of the test charge:

$$\vec{E} = \frac{\vec{F}}{q_0}$$

- Electric field lines provide a means for visualizing the direction and magnitude of electric fields. The electric field vector at any point is tangent to a field line through that point. The density of field lines in any region is proportional to the magnitude of the electric field in that region.
- Field lines originate on positive charge and terminate on negative charge.
- Field due to a point charge:

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r}$$

- The direction is away from the point charge if the charge is positive and toward it if the charge is negative.
- Field due to an electric dipole:

$$E = \frac{1}{2\pi\varepsilon_0}\frac{p}{z^3}$$

- Field due to a continuous charge distribution: treat charge elements as point charges and then summing via integration, the electric field vectors produced by all the charge elements.
- Force on a point charge in an electric field:

$$\vec{F} = q\vec{E}$$

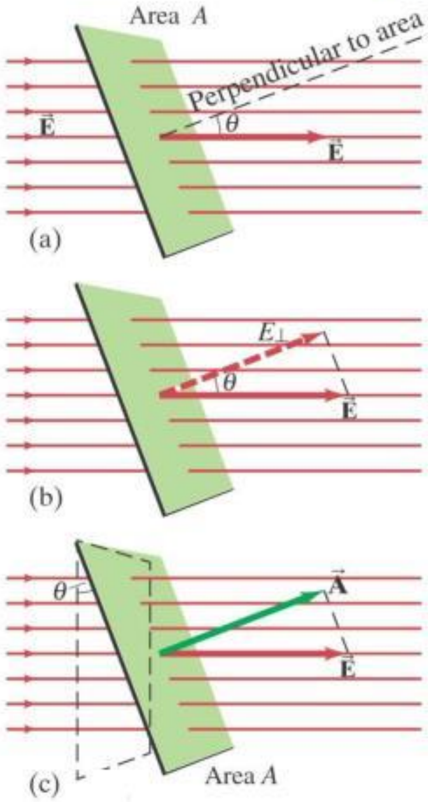
- Dipole in an electric field:
  The field exerts a torque on the dipole

$$\boxed{\vec{\tau} = \vec{p} \times \vec{E}}$$

- The dipole has a potential energy U associated with its orientation in the field

$$\boxed{U = -\vec{p} \cdot \vec{E}}$$

Electric Flux



Electric flux:

$$\Phi_E = E_\perp A = EA_\perp = EA\cos\theta, \qquad [\vec{\mathbf{E}} \text{ uniform}]$$

$$\Phi_E = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}}. \qquad [\vec{\mathbf{E}} \text{ uniform}]$$

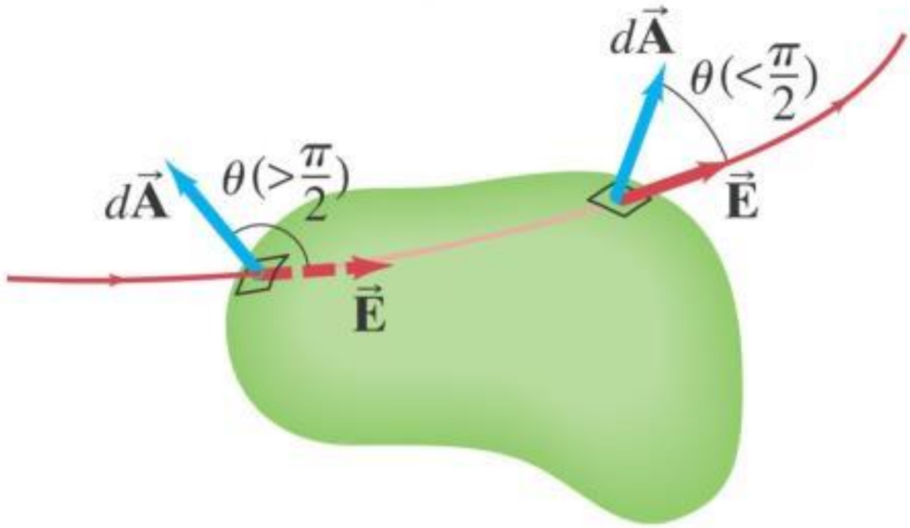
Electric flux through an area is proportional to the total number of field lines crossing the area.

Electric Flux

Flux through a closed surface:

$$\Phi_E \approx \sum_{i=1}^{n} \vec{\mathbf{E}}_i \cdot \Delta\vec{\mathbf{A}}_i .$$

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}.$$



Electric Potential Energy

- The potential energy of the system

$$\boxed{\Delta U = U_f - U_i = -W}$$

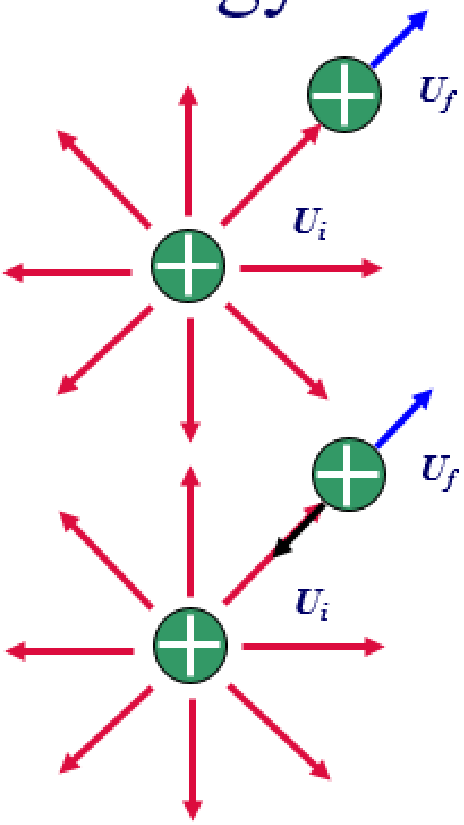
- The work done by the electrostatic force is path independent.
- Work done by a electric force or "field"

$$\boxed{W = \vec{F} \cdot \Delta\vec{r} = q\vec{E} \cdot \Delta\vec{r}}$$

- Work done by an Applied force

$$\boxed{\Delta K = K_f - K_i = W_{app} + W}$$

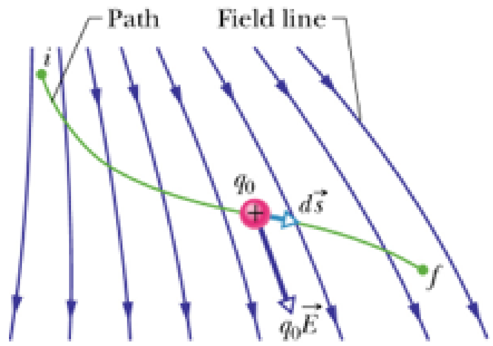
$$\boxed{W_{app} = -W} \qquad \boxed{\Delta U = U_f - U_i = W_{app}}$$



Electric Potential

- The electric potential energy
  - Start $dW = \vec{F} \cdot d\vec{s}$
  - Then $dW = q_0 \vec{E} \cdot d\vec{s}$
  - So $W = q_0 \int_i^f \vec{E} \cdot d\vec{s}$

$$\boxed{\Delta U = U_f - U_i = -W = -q_0 \int_i^f \vec{E} \cdot d\vec{s}}$$



- The electric potential $V = \frac{U}{q}$

$$\Delta V = V_f - V_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q}$$

$$\boxed{\Delta V \equiv \frac{\Delta U}{q_0} = -\int_i^f \vec{E} \cdot d\vec{s}}$$

- Potential difference depends only on the source charge distribution (Consider points i and f without the presence of the test charge;

- The difference in potential energy exists only if a test charge is moved between the points.

Problem 1: What will be the electrostatic force between the two-point charges of charges +2μC and +4μC repel each other with a force of 20N when a charge of −6μC is added to each of them?
Given that,

The first charge, q_1 is +2μC, q_2 is +4μC, q_3 is -6μC.

The electrostatic force in the first case, F is 20 N. $F = k\, q_1 q_2 / r^2$ or $20\text{ N} = k\,(+2\mu C \times +4\mu C) / r^2$ -----(1)

$q_1' = (2 - 6)\ \mu C = -4\ \mu C$; $q_2' = (4 - 6)\ \mu C = -2\ \mu C$

Then, the electrostatic force in this new case is:

$F' = k\, q'_1 q'_2 / r^2 = k\,(-4\ \mu C \times -2\ \mu C) / r^2$ ----- (2)
Now, in order to obtain F', lets divide equation (2) by (1) as,

$F' / 20\text{ N} = [k\,(-4\ \mu C \times -2\ \mu C) / r^2] / [20\text{ N} = k\,(+2\mu C \times +4\mu C) / r^2\,]$

$F' = +20\text{ N}$

Hence, the electrostatic force when third charge is introduced is +20 N.

Summary

- Electric flux is a property of an electric field defined as the number of electric lines of force (or electric field lines) that intersect a given area.
- Electric Potential Energy: a point charge moves from *i* to *f* in an electric field, the change in electric potential energy is
$$\Delta U = U_f - U_i = -W$$
- Electric Potential Difference between two points *i* and *f* in an electric field:
$$\Delta V = V_f - V_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q}$$
- Equipotential surface: the points on it all have the same electric potential. No work is done while moving charge on it. The electric field is always directed perpendicularly to corresponding equipotential surfaces.
- Electric potential: amount of work energy needed per unit of electric charge to move the charge from a reference point to a specific point in an electric field
- Finding V from E: $V(r) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$
- Potential due to point charges:
$$\Delta V \equiv \frac{\Delta U}{q_0} = -\int_i^f \vec{E} \cdot d\vec{s}$$
- Potential due to a collection of point charges:
$$V = \sum_{i=1}^{n} V_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i}$$
- Potential due to a continuous charge distribution:
$$V = \int dV = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}$$
- Potential of a charged conductor is constant everywhere inside the conductor and equal to its value to its value at the surface.
- Calculating E from V:
$$E_s = -\frac{\partial V}{\partial s} \quad E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z}$$
- Electric potential energy of system of point charges:
$$U = q_2 V = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$$